

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

C:\Users\Prashant\anaconda3\lib\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).
from pandas.core import (

```
In [2]: df = pd.read_csv('placement.csv')
```

```
In [3]: df.shape
```

```
Out[3]: (1000, 3)
```

```
In [4]: df.head()
```

```
Out[4]:
```

	cgpa	placement_exam_marks	placed
0	7.19	26.0	1
1	7.46	38.0	1
2	7.54	40.0	1
3	6.42	8.0	1
4	7.23	17.0	0

```
In [14]: plt.figure(figsize=(16,5))
plt.subplot(1,2,1)
sns.distplot(df['cgpa'])

plt.subplot(1,2,2)
sns.distplot(df['placement_exam_marks'])

plt.show()
```

```
C:\Users\Prashant\AppData\Local\Temp\ipykernel_2516\3792736820.py:3: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['cgpa'])
```

```
C:\Users\Prashant\anaconda3\lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
```

```
with pd.option_context('mode.use_inf_as_na', True):
```

```
C:\Users\Prashant\AppData\Local\Temp\ipykernel_2516\3792736820.py:6: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

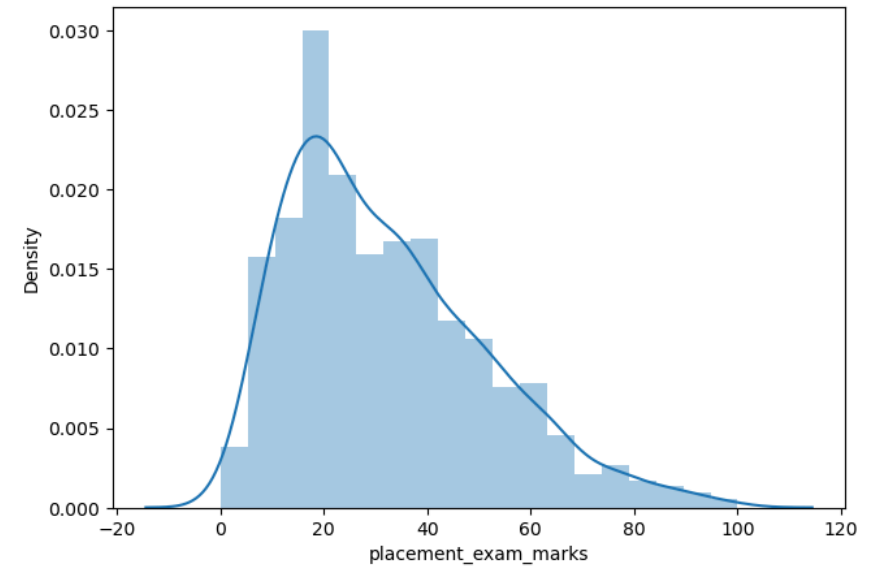
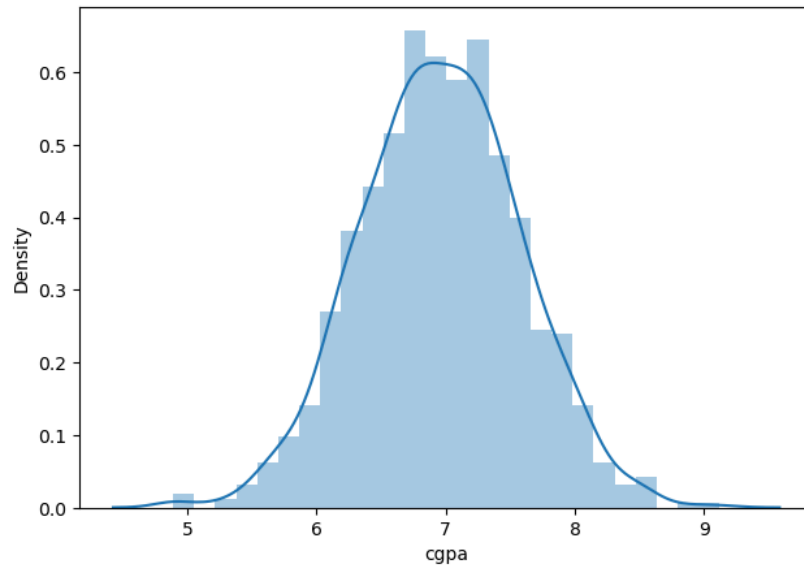
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['placement_exam_marks'])
```

```
C:\Users\Prashant\anaconda3\lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
```

```
with pd.option_context('mode.use_inf_as_na', True):
```

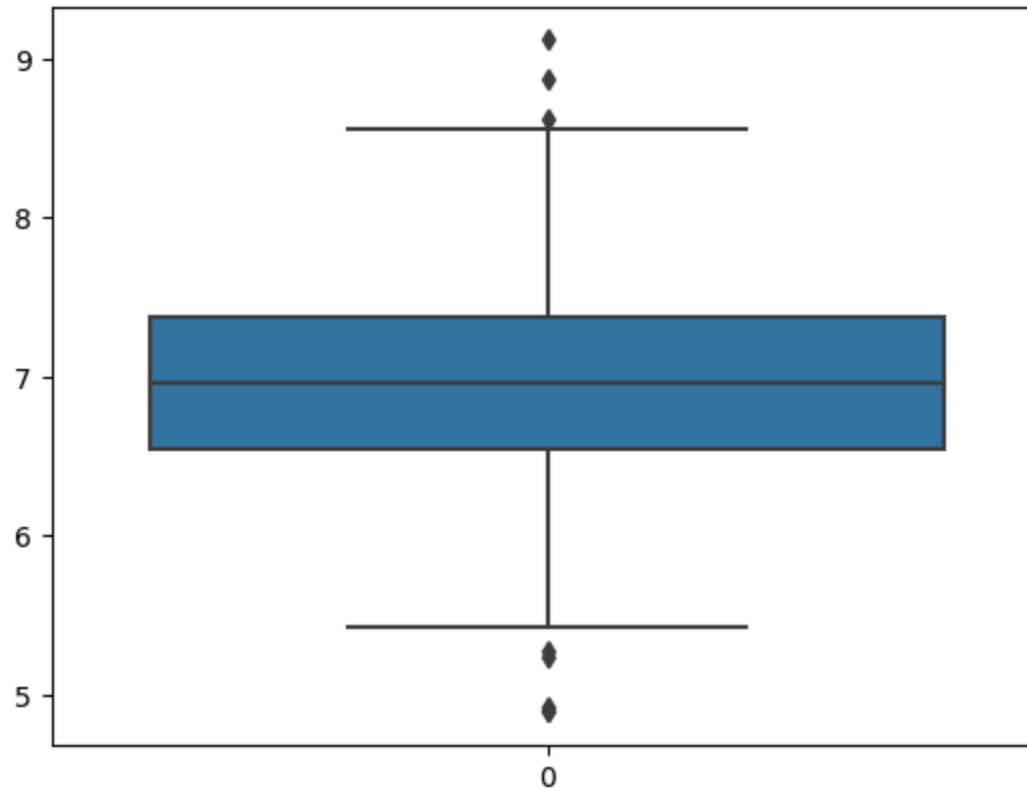


```
In [15]: df['placement_exam_marks'].skew()
```

```
Out[15]: 0.8356419499466834
```

```
In [16]: sns.boxplot(df["cgpa"])
```

```
Out[16]: <Axes: >
```



```
In [17]: # print("Mean value of cgpa",df['cgpa'].mean())
# print("Std value of cgpa",df['cgpa'].std())
# print("Min value of cgpa",df['cgpa'].min())
# print("Max value of cgpa",df['cgpa'].max())
df["cgpa"].describe()
```

```
Out[17]: count    1000.000000
mean         6.961240
std          0.615898
min          4.890000
25%          6.550000
50%          6.960000
75%          7.370000
max          9.120000
Name: cgpa, dtype: float64
```

```
In [18]: # Finding the boundary values
print("Highest allowed",df['cgpa'].mean() + 3*df['cgpa'].std())
print("Lowest allowed",df['cgpa'].mean() - 3*df['cgpa'].std())
```

Highest allowed 8.808933625397168
Lowest allowed 5.113546374602832

```
In [19]: # Finding the outliers  
df[(df['cgpa'] > 8.80) | (df['cgpa'] < 5.11)]
```

```
Out[19]:
```

	cgpa	placement_exam_marks	placed
485	4.92	44.0	1
995	8.87	44.0	1
996	9.12	65.0	1
997	4.89	34.0	0
999	4.90	10.0	1

Trimming

```
In [20]: # Trimming  
  
new_df = df[(df['cgpa'] < 8.80) & (df['cgpa'] > 5.11)]  
new_df
```

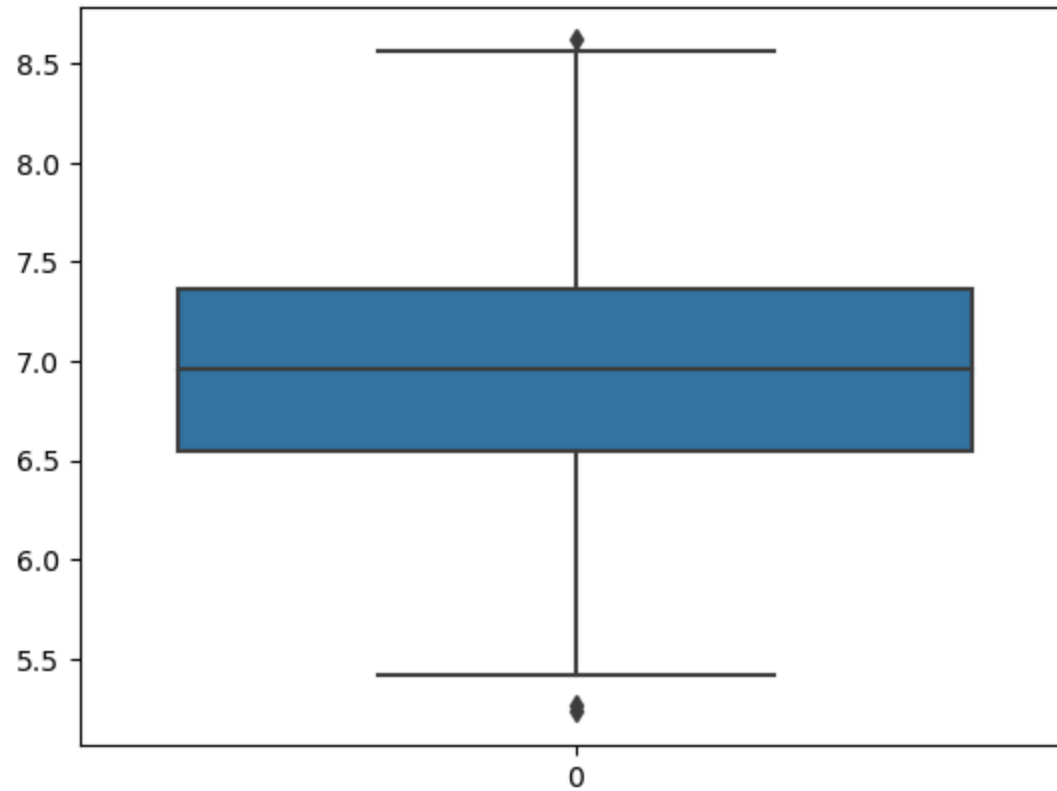
Out[20]:

	cgpa	placement_exam_marks	placed
0	7.19	26.0	1
1	7.46	38.0	1
2	7.54	40.0	1
3	6.42	8.0	1
4	7.23	17.0	0
...	...	...	...
991	7.04	57.0	0
992	6.26	12.0	0
993	6.73	21.0	1
994	6.48	63.0	0
998	8.62	46.0	1

995 rows × 3 columns

In [21]: `sns.boxplot(new_df["cgpa"])`

Out[21]: <Axes: >



```
In [22]: # Approach 2  
  
# Calculating the Zscore -3 se 3  
  
df['cgpa_zscore'] = (df['cgpa'] - df['cgpa'].mean())/df['cgpa'].std()
```

```
In [23]: df.head()
```

Out[23]:

	cgpa	placement_exam_marks	placed	cgpa_zscore
0	7.19	26.0	1	0.371425
1	7.46	38.0	1	0.809810
2	7.54	40.0	1	0.939701
3	6.42	8.0	1	-0.878782
4	7.23	17.0	0	0.436371

In [24]: `df[df['cgpa_zscore'] > 3] # -3 +3`

Out[24]:

	cgpa	placement_exam_marks	placed	cgpa_zscore
995	8.87	44.0	1	3.099150
996	9.12	65.0	1	3.505062

In [25]: `df[df['cgpa_zscore'] < -3]`

Out[25]:

	cgpa	placement_exam_marks	placed	cgpa_zscore
485	4.92	44.0	1	-3.314251
997	4.89	34.0	0	-3.362960
999	4.90	10.0	1	-3.346724

In [26]: `df[(df['cgpa_zscore'] > 3) | (df['cgpa_zscore'] < -3)]`

Out[26]:

	cgpa	placement_exam_marks	placed	cgpa_zscore
485	4.92	44.0	1	-3.314251
995	8.87	44.0	1	3.099150
996	9.12	65.0	1	3.505062
997	4.89	34.0	0	-3.362960
999	4.90	10.0	1	-3.346724

```
In [27]: # Trimming  
new_df = df[(df['cgpa_zscore'] < 3) & (df['cgpa_zscore'] > -3)]
```

```
In [30]: new_df.describe()
```

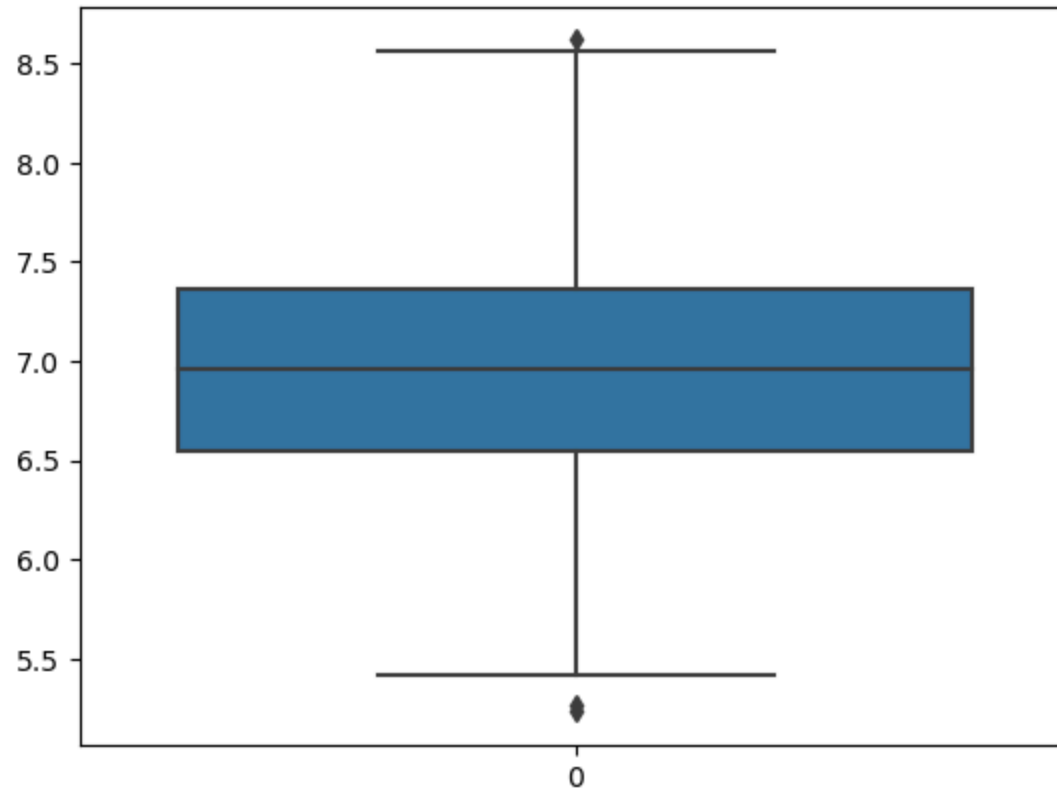
```
Out[30]:
```

	cgpa	placement_exam_marks	placed	cgpa_zscore
count	995.000000	995.000000	995.000000	995.000000
mean	6.963357	32.188945	0.487437	0.003437
std	0.600082	19.130298	0.500094	0.974320
min	5.230000	0.000000	0.000000	-2.810921
25%	6.550000	17.000000	0.000000	-0.667708
50%	6.960000	28.000000	0.000000	-0.002013
75%	7.365000	44.000000	1.000000	0.655563
max	8.620000	100.000000	1.000000	2.693239

Capping

```
In [29]: sns.boxplot(new_df["cgpa"])
```

```
Out[29]: <Axes: >
```



```
In [69]: upper_limit = df['cgpa'].mean() + 3*df['cgpa'].std()  
lower_limit = df['cgpa'].mean() - 3*df['cgpa'].std()
```

```
In [71]: lower_limit
```

```
Out[71]: 5.113546374602842
```

```
In [72]: df['cgpa'] = np.where(  
    df['cgpa'] > upper_limit,  
    upper_limit,  
    np.where(  
        df['cgpa'] < lower_limit,  
        lower_limit,  
        df['cgpa']  
    )  
)
```

```
In [73]: df.shape
```

```
Out[73]: (1000, 4)
```

```
In [74]: df['cgpa'].describe()
```

```
Out[74]: count    1000.000000  
mean         6.961499  
std          0.612688  
min          5.113546  
25%          6.550000  
50%          6.960000  
75%          7.370000  
max          8.808934  
Name: cgpa, dtype: float64
```

```
In [ ]:
```